Double Q-Learning and Double DQN

Deep Reinforcement Learning

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Contents

1 Maximum Estimation and Maximization Bias

Proposition 1.1. *Let $X_1, \ldots, X_m$ be integrable random variables. Then*

$$\boxed{\mathbb{E}\left[\max_{1\le i\le m} X_i\right] \ge \max_{1\le i\le m} \mathbb{E}[X_i]}.$$

Proof. For every $i \in \{1, \ldots, m\}$,

$$\max_{1\le j\le m} X_j \ge X_i.$$

Taking expectations,

$$\mathbb{E}\left[\max_{1\le j\le m} X_j\right] \ge \mathbb{E}[X_i], \qquad i = 1, \ldots, m.$$

Since this inequality holds for every i,

$$\mathbb{E}\left[\max_{1\le i\le m} X_i\right] \ge \max_{1\le i\le m} \mathbb{E}[X_i].$$

$\square$

1.1 Example: Estimating the Maximum of Two Population Means

Consider two mutually independent random samples

$$X_{1,1}, \ldots, X_{1,N} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_1, \sigma_1^2), \qquad X_{2,1}, \ldots, X_{2,N} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_2, \sigma_2^2),$$

where the population means μ_1 and μ_2 are unknown and $\sigma_1^2, \sigma_2^2 > 0$.

The unknown quantity of interest is

$$\mu_{\max} := \max\{\mu_1, \mu_2\}.$$

Define the sample means

$$\overline{X}_1 := \frac{1}{N}\sum_{j=1}^{N} X_{1,j}, \qquad \overline{X}_2 := \frac{1}{N}\sum_{j=1}^{N} X_{2,j}.$$

Their sampling distributions are

$$\overline{X}_1 \sim \mathcal{N}\left(\mu_1, \frac{\sigma_1^2}{N}\right), \qquad \overline{X}_2 \sim \mathcal{N}\left(\mu_2, \frac{\sigma_2^2}{N}\right).$$

Therefore,

$$\mathbb{E}[\overline{X}_1] = \mu_1, \qquad \mathbb{E}[\overline{X}_2] = \mu_2,$$

so both sample means are unbiased estimators of their corresponding population means.

A natural estimator of $\mu_{\max}$ is

$$\widehat{\mu}_{\max} := \max\{\overline{X}_1, \overline{X}_2\}.$$

Applying the proposition,

$$
\begin{aligned}
\mathbb{E}[\widehat{\mu}_{\max}] &= \mathbb{E}\left[\max\{\overline{X}_1, \overline{X}_2\}\right] \\
&\geq \max\left\{\mathbb{E}[\overline{X}_1], \mathbb{E}[\overline{X}_2]\right\} \\
&= \max\{\mu_1, \mu_2\} \\
&= \mu_{\max}.
\end{aligned}
$$

Hence,

$$\boxed{\mathrm{Bias}(\widehat{\mu}_{\max}) := \mathbb{E}[\widehat{\mu}_{\max}] - \mu_{\max} \geq 0}.$$

Thus, although $\overline{X}_1$ and $\overline{X}_2$ are individually unbiased, their maximum is an upward-biased estimator of the maximum population mean. Because the two sample means are independent, nondegenerate normal random variables, their distributions overlap, and the inequality is strict for every finite N:

$$\mathbb{E}[\widehat{\mu}_{\max}] > \mu_{\max}.$$

The expectation $\mathbb{E}[\widehat{\mu}_{\max}]$ represents the average estimate over repeated realizations of the experiment. Thus, positive bias means that the estimator is greater than the true unknown value $\mu_{\max}$ on average; an individual realization may still lie either above or below $\mu_{\max}$.

2 Maximization Bias in Q-Learning

Q-learning approximates the Bellman optimality backup using both a sampled environment transition and estimated action values. These introduce two different sources of randomness. We first show that sampling the environment transition alone does not introduce maximization bias, and then isolate the bias that appears when the maximum is taken over estimated action values.

2.1 Bellman Optimality Backup

The optimal state-action value function satisfies

$$\boxed{Q^*(s,a) = \mathbb{E}_{(S_{t+1},R_{t+1})\sim p(\cdot,\cdot|s,a)}\left[R_{t+1} + \gamma \max_{a'\in\mathcal{A}} Q^*(S_{t+1},a')\right].} \tag{1}$$

Suppose for the moment that Q^* is known exactly. From one sampled environment transition

$$(S_{t+1}, R_{t+1}),$$

define the sampled Bellman target

$$Y = R_{t+1} + \gamma \max_{a'\in\mathcal{A}} Q^*(S_{t+1},a'). \tag{2}$$

Conditioned on $S_t = s$ and $A_t = a$,

$$
\begin{aligned}
\mathbb{E}\left[Y \mid S_t = s, A_t = a\right] &= \mathbb{E}_{(S_{t+1},R_{t+1})\sim p(\cdot,\cdot|s,a)}\left[R_{t+1} + \gamma \max_{a'\in\mathcal{A}} Q^*(S_{t+1},a')\right] \\
&= Q^*(s,a).
\end{aligned} \tag{3}
$$

Thus Y is an unbiased one-sample estimate of the Bellman optimality backup. Sampling the environment introduces variability in the target, but not maximization bias: for every realized next state, the quantities $Q^*(S_{t+1},a')$ being maximized are assumed to be known exactly.

2.2 Maximization over Estimated Action Values

In Q-learning, the optimal action values are not known exactly and must instead be estimated. To isolate the effect of this estimation error, fix one realized environment transition

$$(S_{t+1}, R_{t+1}) = (s', r).$$

The next state s' and immediate reward r are now fixed; the remaining randomness comes from the estimated action values.

Let the finite action space be

$$\mathcal{A} = \{a_1, \ldots, a_m\}.$$

For the fixed state s', suppose that each

$$\widehat{Q}(s', a_i)$$

is an unbiased estimator of the corresponding optimal action value:

$$\mathbb{E}\left[\widehat{Q}(s', a_i)\right] = Q^*(s', a_i), \qquad i = 1, \ldots, m.$$

This assumption isolates the effect of maximization: even if the individual action-value estimates are unbiased, taking their maximum can introduce positive bias. No independence among the estimators is required.

Q-learning estimates the optimal value at s' using

$$\max_{1 \le i \le m} \widehat{Q}(s', a_i).$$

Applying the maximization result from the preceding section,

$$\begin{aligned} \mathbb{E}\left[\max_{1 \le i \le m} \widehat{Q}(s', a_i)\right] &\ge \max_{1 \le i \le m} \mathbb{E}\left[\widehat{Q}(s', a_i)\right] \\ &= \max_{1 \le i \le m} Q^*(s', a_i). \end{aligned} \tag{4}$$

Hence,

$$\boxed{\mathbb{E}\left[\max_{a' \in \mathcal{A}} \widehat{Q}(s', a')\right] \ge \max_{a' \in \mathcal{A}} Q^*(s', a').} \tag{5}$$

For the same fixed transition (s', r), the estimated Q-learning target is

$$\widehat{Y} = r + \gamma \max_{a' \in \mathcal{A}} \widehat{Q}(s', a'),$$

and therefore

$$\boxed{\mathbb{E}\left[\widehat{Y} \mid S_{t+1} = s', R_{t+1} = r\right] \ge r + \gamma \max_{a' \in \mathcal{A}} Q^*(s', a').} \tag{6}$$

Thus the upward bias is not caused by sampling the environment transition. It appears when Q-learning maximizes over noisy estimates of the action values, even when each estimate is individually unbiased.

3 Double Deep Q-Network

The maximization bias derived above arises because Q-learning uses the same action-value estimates both to select the maximizing action and to evaluate the value of that action.

To make these two roles explicit, the Q-learning target can be written as

$$a_i^* = \arg\max_{a' \in \mathcal{A}} \widehat{Q}(S_{i+1}, a'),$$

followed by

$$Y_i = R_{i+1} + \gamma \widehat{Q}(S_{i+1}, a_i^*).$$

The action a_i^* is therefore selected because its estimated value is the largest. When the estimates contain error, actions with positive estimation errors are more likely to win this maximization. Evaluating the selected action with the same estimates then preserves that optimistic error in the target.

The central idea of Double Q-learning is to separate these two operations. Suppose two action-value estimators are available,

$$\widehat{Q}^A(s, a), \qquad \widehat{Q}^B(s, a).$$

One estimator is used to select the greedy action,

$$a_i^* = \arg\max_{a' \in \mathcal{A}} \widehat{Q}^A(S_{i+1}, a'),$$

while the second estimator is used to evaluate that selected action,

$$Y_i = R_{i+1} + \gamma \widehat{Q}^B(S_{i+1}, a_i^*).$$

The estimator that selects the action may still select an action because of a positive estimation error, but that same error is no longer automatically used to evaluate the action. This decoupling reduces the positive maximization bias caused by using the same estimates for both selection and evaluation.

In the original Double Q-learning algorithm, two value functions are learned and their selection and evaluation roles are exchanged symmetrically across updates. DQN already maintains two sets of parameters: the online network w and the target network $\bar{w}$. Double DQN therefore adapts the Double Q-learning idea without introducing another pair of networks.

DQN uses the target network for both action selection and action evaluation:

$$a_i^* = \arg\max_{a' \in \mathcal{A}} \hat{q}(S_{i+1}, a', \bar{w}), \qquad Y_i = R_{i+1} + \gamma \hat{q}(S_{i+1}, a_i^*, \bar{w}).$$

Double DQN instead uses the online network to select the maximizing action,

$$\boxed{a_i^* = \arg\max_{a' \in \mathcal{A}} \hat{q}(S_{i+1}, a', w),} \tag{7}$$

and the target network to evaluate it,

$$\boxed{Y_i = R_{i+1} + \gamma \hat{q}(S_{i+1}, a_i^*, \bar{w}).} \tag{8}$$

Thus,

$$\boxed{\text{DQN: selection with } \bar{w}, \text{ evaluation with } \bar{w}}$$

is replaced by

$$\boxed{\text{Double DQN: selection with } w, \text{ evaluation with } \bar{w}.}$$

All remaining components of DQN—experience replay, online-network regression, the behaviour policy, and target-network refresh—remain unchanged.

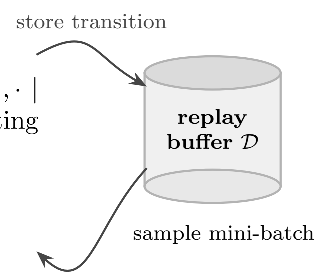
Initialize online parameters w, target parameters $\bar{w} \leftarrow w$, replay buffer $\mathcal{D} \leftarrow \varnothing$.

for $m = 1, \ldots, M$

  $\bar{w} \leftarrow w$ *(refresh the target)*

  for $k = 1, \ldots, K$

    Sample $A_t \sim \pi_{b,t}(\cdot \mid S_t)$, then $(S_{t+1}, R_{t+1}) \sim p(\cdot, \cdot \mid S_t, A_t)$, and append $(S_t, A_t, R_{t+1}, S_{t+1})$ to $\mathcal{D}$, evicting the oldest transition if $|\mathcal{D}| > C$.



    for $u = 1, \ldots, U$

1. Sample a mini-batch

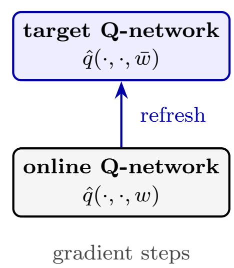
$$\mathcal{B}_t = \{(S_i, A_i, R_{i+1}, S_{i+1})\}_{i=1}^{B} \sim \text{Uniform}(\mathcal{D}).$$

2. Select the maximizing action with the *online* network:

$$a_i^* = \arg\max_{a' \in \mathcal{A}} \hat{q}(S_{i+1}, a', w).$$

Evaluate that action with the *target* network:

$$Y_i = R_{i+1} + \gamma \hat{q}(S_{i+1}, a_i^*, \bar{w}).$$



3. Update the *online* network:

$$w \leftarrow w + \frac{\alpha}{B} \sum_{i=1}^{B} [Y_i - \hat{q}(S_i, A_i, w)] \nabla_w \hat{q}(S_i, A_i, w).$$

Figure 1: Double DQN retains the same generalized Q-learning structure as DQN, with only the bootstrapped target changed: the online network selects the maximizing action, while the target network evaluates it.